ADITYA UNIVERSITY

(Formerly known as Aditya Engineering College (A))

000386

PART-I

S.No: 421926

Programe	: B.TECH
Hall Ticket No:	21A91A0341
Name	: SOMESWARAPU SAI CHANDU
Examination	: III SEMESTER SUPPLEMENTARY (AR20)
Month-Year	: Nov/Dec 2025
Branch	: MECHANICAL ENGINEERING
Course Code	: 201BS3T12
Course Name	: Integral transforms & applications of Partial Differential Equations
Date of Exam	: 18-11-2025

J. Pavan.

Signature of the Controller of Examination

S. Ch 18-11-2025

Signature of the Student with date

M. L. Chithra 18/11/25

Signature of the Invigilator with date

Serial No. Last Page Written	34

INSTRUCTIONS TO STUDENTS

1. The Answer Booklet contains 36 pages Ensure all the 36 pages are in proper order.
2. Student must verify the details of particulars in the PART - 1 i.e Name, Hall Ticket No., Examination, Course Name etc.,
3. In case of any deviation in the above or if the PART-1 is torn / damaged, report to the invigilator and return the damaged booklet.
4. You are prohibited from writing on or tampering the PART-1 except affixing the signature and serial number of last page written in the space provided.
5. Student is prohibited from:
   i. Writing their Hall Ticket No. and name in any part of the answer booklet.
   ii. Addressing the examiner in any manner whatsoever in the answer booklet. If they do so, their script will not be valued.
   iii. Writing religious symbols.
   iv. Either seeking or providing any assistance to the fellow students in the examinations.
   v. Possessing a manuscript or a printed matter, in any form, in the examination hall.
   vi. Bringing Mobile Phones / Cameras/ Bluetooth Devices / Programmable calculators etc.
   vii. Violation of the above mentioned instructions will be viewed as a case of malpractice, which is punishable offence.
6. Before beginning to answer any question, the students should write the correct number of that question in the margin provided. Answers written at different places for the same question will not be valued.
7. Students should write the answers, within the margins provided on both sides of the paper and on all the lines of each page. It is not necessary to begin each answer in a fresh page. Answers must be legibly written with blue or black pen.
8. Do not write anything except Question Number in the margin.
9. No loose sheets of papers will be allowed in the examination hall. No paper must be detached from or attached to the answer booklet except graph sheet.
10. Strike off all unused pages.
11. **NO ADDITIONAL ANSWER BOOKLETS/SHEETS WILL BE SUPPLIED.**

SPACE FOR ROUGH WORK

$\frac{2}{\pi}\int_0^2 f(x)\,dx$

$\frac{1}{2a}(\sin at \cdot u - (\cos at - u)/2a)$

$\frac{2}{s^3}\left(\frac{1}{s+3}\right)\left(\frac{2}{s^2+4}\right), \frac{2}{s^3}$

$y(x,t) = \sum C_n \frac{n\pi ct}{l} \cdot D_n \frac{n\pi ct}{l}\frac{\sin n\pi}{l}$

$L(e^{at})$

$\frac{b}{s^2+b^2}$ $\frac{2}{s^2+(2)^2}$ $L(e^{at}t^n)$

$\frac{2}{s^2+4^2}$

$L^{-1}(F(s)\ \bar{g}(s))\frac{1}{2a} =$

$x'(x)\cdot T(t) = 2x(x)\cdot T'(t) + X(x)\cdot T(t)$

$\sin bt$

$\cos bt$

$L\{t^2 e^{-3t} \sin 2t\}$ $\frac{1}{s-a}$

$\sinh bt$

$\left(\frac{1}{s+3}\right)$ $\frac{du}{dx} = x'(x)\cdot T(t)$

$\cosh bt$

$L\{t^2\}$

$L\{e^{-3t}\}$ $\frac{du}{dt} = x(x)\cdot T'(t)$

$x'(x)\cdot = 2x(x)\cdot T'(t) + X(x)$ $L\{t^2\}$

$L\cdot\{\sin 2t\}$

$x'(x)T(t) = 2x(x)\left[T'(t) + T(t)\right]$. $\frac{n!}{(s)^{n+1}}$

$\frac{x'(x)}{x(x)} = \frac{2\left[T'(t) + T(t)\right]}{T(t)}$ $\left[\frac{1}{s^2}\right]$

$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}$

Q.No.

9a,

Given that,

$$\frac{du}{dx} = 2\frac{du}{dt} + u$$

where $u(x,0) = 6e^{-3x}$

from given equation $\frac{du}{dx} = 2\frac{du}{dt} + u$

Where,

$$u = X(x) \cdot T(t) \quad -(1)$$

$$\frac{du}{dx} = X'(x) \cdot T(t) \quad -(2)$$

$$\frac{du}{dt} = X(x) \cdot T'(t) \cdot \quad -(3)$$

NOW, Subtuting the values in the given equation. by using Method of Separation of variables from Eq-(1) given.

$$\frac{du}{dx} = 2\frac{du}{dt} + u$$

Q.No.

$$\Rightarrow X'(x) \cdot T(t) = 2\left(X(x) \cdot T'(t)\right) + X(x) \cdot T(t).$$

$$X'(x) \cdot T(t) = 2X(x) \cdot T'(t) + X(x) \cdot T(t).$$

$$\Rightarrow X'(x) \cdot T(t) = 2X(x)\left[T'(t) + T(t)\right]$$

$$\Rightarrow \frac{X'(x)}{X(x)} = \frac{2\left(T'(t) + T(t)\right]}{T(t).} = k \text{ Say.}$$

∴ According to the Method of Separation of the variables.

$\frac{X'(x)}{X(x)} = k.$	$\frac{2\left[T'(t) + T(t)\right]}{T(t)} = k.$
$X'(x) = kX(x).$	$2(T'(t) + T(t) = kT(t)$
$X'(x) - kX(x) = 0$	$2T'(t) + T(t) - kT(t) = 0$
$(D - k)X = 0$	$2T'(t) - T(t) + kT(t) = 0$

Q.No.

by auxillary Equn

$M - k = k.$

$\Rightarrow M = k^{x}$

$\boxed{\therefore X(x) = c_1 e^{kx}}$

$2D + k - 1 = 0$

by auxillary equn.

$MD + k - 1 = 0$

$\boxed{\Rightarrow M = \frac{k-1}{2}}$

$\boxed{T(t) = c_2 e^{\frac{k-1}{2}}}$

from the eq (1).

$u = X(x) \cdot T(t)$

$u = c_1 e^{kx} \cdot c_2 e^{\frac{k-1}{2}}$

where $u(x,0) = 6e^{-3x}.$

where $\boxed{t = 0}$

$u = c_1 c_2 e^{kx}.$

$\Rightarrow 6e^{-3x} =$

$c_1 c_2 = 6$

$k = -3.$

Q.No.

$\therefore \boxed{u(x,0) = 6e^{-3x-2t}}$

Q.No.

9b,

The possible solutions of the wave equation.

The one diemensional wave equation

$$\frac{d^2y}{dt^2} = c^2 . \frac{d^2y}{dx^2}$$

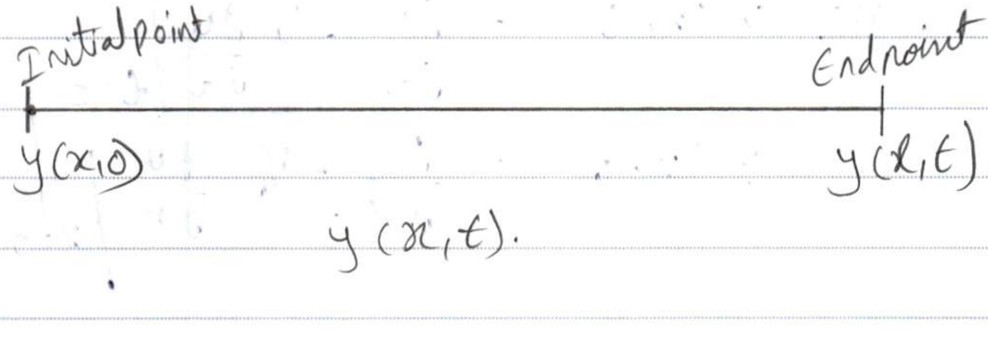
Here,

$$c^2 = T/m$$

where,

T = Tensile strength on the length of the string.

m = Mass of the string which is unit per length of the string.



Q.No.

where,

from the one diemensiional wave Eq

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}.$$

The possible solutions of the wave Equation.

i, $y(x,0) = 0 \Rightarrow 0 \; \forall t.$

where the initial strech point is always zero (0).

ii, $y(l,t) = 0 \Rightarrow 0 \; \forall t.$

where the end point of struch is also always zero (0).

iii, $y(x,t) = \frac{dy}{dt} = \text{if } \left.\frac{dy}{dt}\right|_{t=0} = 0 \leq x \leq t$

$y(x,t) = \frac{dy}{dx} = \text{if } \left.\frac{dy}{dx}\right|_{x=0} = 0 \leq x \leq t.$

Q.No.

Where the Solution of the wave equation is

$$\Rightarrow \boxed{y(x,t) = \sum_{n=1}^{\infty}\left(C_n \frac{n\pi ct}{l} \cdot D_n \frac{n\pi ct}{l}\right) \sin\frac{n\pi x}{l}.}$$

Q.No.

6a)

laplace transforms.

The given laplace transform.

$L\{t^2 e^{-3t} \sin 2t\}$.

The standard formulars of laplace transforms.

① $L(e^{at}) = \frac{1}{s-a}$

② $L(e^{at} t^n) = \frac{n!}{(s-a)^{n+1}}$

③ $L(e^{at} \sin bt) = \frac{b}{(s-a)^2+b^2}$

④ $L(e^{at} \cos bt) = \frac{s-a}{(s-a)^2+b^2}$.

⑤ $L(e^{at} \sinh bt) = \frac{b}{(s-a)^2-b^2}$

⑥ $L(e^{at} \cosh bt) = \frac{s-a}{(s-a)^2-b^2}$

Q.No.

$$L(1) = \frac{1}{s}$$

$$L(t^n) = \frac{n!}{(s)^{n+1}}$$

$$L(\sin bt) = \frac{b}{s^2+b^2}$$

$$L(\cos bt) = \frac{s}{s^2+b^2}$$

$$L(\sinh bt) = \frac{b}{s^2-b^2}$$

$$L(\cosh bt) = \frac{s}{s^2-b^2}$$

Trignometric irregulaties

1, $2\cos A \cos B = \cos(A+B) + \cos(A-B)$

2, $2\sin A \sin B = \cos(A-B) - \cos(A+B)$

3, $2\cos A \sin B = \sin(A+B) + \sin(A-B)$

4, $2\sin A \cos B = \sin(A+B) - \sin(A-B)$.

5, $\cos 3t = 4\cos^3 t - \cos 3t$

$$\Rightarrow \cos^3 t = \frac{\cos 3t - 3\cos t}{4}$$

Q.No.

6, $\sin 3t = 3\sin t - 4\sin 3t$

$$\Rightarrow \sin^3 t = \frac{3\sin t - \sin 3t}{4.}$$

7, ~~Sin~~ $\cos^2 t = \frac{1 + \cos 2at}{2.}$

8, $\sin^2 t = \frac{1 - \cos 2at}{2.}$

The given laplace transform.

$L\{t^2 e^{-3t} \sin 2t\}$. —— (1).

Now,

applying laplace transform to the given equation.

$\Rightarrow = L\{t^2\}$

$$\boxed{= \therefore L(t^n) = \frac{n!}{(s)^{n+1}}}$$

$$= \frac{2!}{(s)^{2+1}}$$

$$\boxed{= \frac{2}{s^3}}$$

Q.No.

$\Rightarrow L\{e^{-3t}\}$

$\boxed{= \therefore L(e^{at}) = \frac{1}{s-a}}$

$\boxed{= \frac{1}{s+3}}$

$\Rightarrow L\{\sin 2t\}$

$= \boxed{L(\sin bt) = \frac{b}{s^2+b^2}}$

where, $\boxed{b=2}$

$\Rightarrow = \frac{2}{s^2+(2)^2}$

$\boxed{= \frac{2}{s^2+4}}$

from Eq ①

$L\{t^2 e^{-3t} \sin 2t\}$.

$\Rightarrow L\left\{\frac{2}{s^3}\left(\frac{1}{s+3}\right)\left(\frac{2}{s^2+4}\right)\right\}$.

Q.No.

$$\Rightarrow = \frac{2}{s^3}\left(\frac{1}{s+3}\right)\left(\frac{2}{s^2+4}\right).$$

$$= \frac{2}{s^3}\left(\frac{2}{s^3+12}\right)$$

$$\Rightarrow \frac{4}{s^6+12}$$

∴ The required Solution is

$$\Rightarrow \boxed{\frac{4}{s^6+12}}$$

Q.No.

6b)

Given that,

evaluate $\int_0^\infty \frac{\cos 6t - \cos 4t}{t}$.

Here,

By using laplace transforms

$\Rightarrow L\left(\frac{\cos 6t}{t}\right) - L\left(\frac{\cos 4t}{t}\right)$

when by using the trigonometric function

$\Rightarrow \frac{1}{2}\left[\frac{2(\cos 6t - \cos 4t)}{t}\right]$

Where, $\boxed{2\cos A \cos B = \cos(A+B) + \cos(A-B)}$

$\Rightarrow = \frac{1}{2}\left[\frac{\cos(6+4) + \cos(6-4)}{t}\right]$

$\Rightarrow = \frac{1t}{2} \left[\cos(6+4) + \cos(6-4)\right]$

Q.No.

$= \frac{t}{2} [\cos(10) + \cos(2)]$

$= \frac{t}{2} [\cos 12]$

$= \frac{1}{2} \cos 12t$

$= \boxed{\cos bt = \frac{s}{s^2+b^2}}$

$= \frac{s}{s^2+(12)^2}$

$= \frac{s}{s^2+144} =$

$\Rightarrow \boxed{\frac{s(t)}{2s^2+144}}$

Q.No.

7b,

given that,

$$L^{-1}\left\{\frac{s^2}{s^2+4(s^2+9)}\right\} \text{ — Eq ①.}$$

By using convolution theorem.

$L^{-1}\{F(t)\}$ $L^{-1}\{\theta(t)\}$

$$L^{-1}(\bar{F}(t)\cdot\bar{g}(t)) = f(s)\cdot g(s).$$

$$\therefore F(s)\cdot g(s) = f(t) * g(t).$$

According to the convolution theorem.

$$\boxed{L(F(s) * g(s)) = \int_{-0}^{t} f(u)\cdot g(t-u)\,du}$$

Q.No.

from the given equation.

$$f(s) = \left[\frac{s^2}{s^2+4}\right] \qquad g(s) = \left[\frac{s^2}{s^2+9}\right]$$

where,

$$f(s) * g(s) = f(t) \cdot g(t).$$

$\therefore$ $f(t) =$ $L\left[\frac{s^2}{s^2+4}\right]$. $\qquad$ $g(t) =$ $L\left[\frac{s^2}{s^2+9}\right]$.

$f(t) = \cos at \qquad g(t) = \frac{1}{a} \cos at$

$$\Rightarrow f(s) \times g(s) = \int_0^t (\cos at \cdot u \cdot \frac{1}{a} \cos at - u)\, du$$

$$= \frac{1}{2a} \left[2 \cos at \cdot u - \cos at - u \right) du.$$

Q.No.

$$= \frac{1}{2a} \left[2 \cos \underline{au} . \cos \underline{(at-u)} \right] du.$$

$A \qquad B$

when, $2 \cos A \cos B = \cos(A+B) + \cos(A-B)$

$$= \frac{1}{2a} \left[\cos au + at - u + \cos au - at - u \right] du.$$

$$= \frac{1}{2a} \left[\cos au . - u \frac{\cos (at-u)}{2a} \right) du.$$

$$= \frac{1}{2a} \left[u + \frac{\sin at - u}{2a} - u - \frac{\cos at - u}{2a} \right) du$$

$$= \frac{1}{2a} \left[t + \frac{\sin at - u}{2a} - \frac{\cos at - u}{2a} \right]$$

$$= \boxed{\frac{t}{2a} - \sin at}$$

Q.No.

1a,

Given that

$$f(x) = \begin{cases} x & \text{if } 0 \le x \le 3 \\ 6-x & 3 \le x \le 6 \end{cases}$$

The Fourier series expansion

$$\cancel{f(x) = \frac{a_0}{2} \int_0^{\pi} \cos nx + \sin nx}$$

$$\boxed{f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \int_0^{\pi} a_n \cos nx + b_n \sin nx}$$

where,

$$a_0 = \frac{1}{\pi} \int_0^{\pi} f(x)\,dx.$$

$$a_n = \frac{1}{\pi} \int_0^{\pi} f(x) \cos nx\,dx.$$

Q.No.

$$b_n = \frac{1}{\pi}\int_0^{\pi} f(x)\sin x\,dx.$$

To find a_0

$$a^0 = \frac{1}{\pi}\int_0^{\pi} f(x)\,dx$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{0} f(x)\,dx + \int_0^{\pi} f(x)\,dx.$$

$$a_0 = \frac{1}{\pi}\left\{\int_{-\pi}^{0}\left(1+\frac{2x}{\pi}\right)dx + \int_0^{\pi}\left(1-\frac{2x}{\pi}\right)dx.\right.$$

$$= \frac{1}{\pi}\left[x+\frac{2}{\pi}\cdot\frac{x^2}{2}\right]_{-\pi}^{0} + \left[x-\frac{2}{\pi}\cdot\frac{x^2}{2}\right]_0^{\pi}$$

$$= \frac{1}{\pi}\left[(0)\left(-\pi+\frac{\pi^2}{\pi}\right]_{-\pi}^{0} + \left[\pi - \frac{\pi^2}{\pi} - (0)\right]_0^{\pi}$$

$$= \frac{1}{\pi}\left[\pi - \pi + \pi - \pi\right]$$

Q.No.

$$= \frac{1}{\pi}[0]$$

$$= 0$$

$$a_0 = 0.$$

<u>To find a_n</u>

$$a_n = \frac{1}{\pi}\int_{-\pi}^{0}\left(1+\frac{2x}{\pi}\right)\cos nx\,dx + \int_{0}^{\pi}\left(1-\frac{2x}{\pi}\right)\cos nx\,dx$$

$$= \frac{1}{\pi}\left[\left(1+\frac{2x}{\pi}\right)\left(\frac{\sin nx}{n}\right) - \left(\frac{2}{\pi}\right)\left(\frac{1}{n} - \frac{\cos nx}{n}\right)\right]_{-\pi}^{0}$$

$$+ \left[\left(1-\frac{2x}{\pi}\right)\left(\frac{\sin nx}{n}\right) - \left(\frac{-2}{\pi}\right)\left(\frac{1}{n} - \frac{\cos nx}{n}\right)\right]_{0}^{\pi}$$

Q.No.

$$= \frac{1}{\pi}\left[\left(\frac{2}{\pi n^2}\right) - \left(\frac{2}{\pi n^2}\frac{1}{n}(-1)^n\right) + \frac{-2}{\pi n^2}\left[\frac{1}{n}(-1)\right]^2 \frac{-2}{\pi n^2}\right]$$

$$= \frac{1}{\pi}\left[\frac{2}{\pi n^2} - \frac{2}{\pi n^2}\frac{1}{n}(-1)^n - \frac{2}{\pi n^2}\frac{1}{n}(-1)^2 + \frac{2}{\pi n^2}\right]$$

$$= \frac{1}{\pi}\left[\frac{4}{\pi n^2} - \frac{4}{\pi n^2}(-1)^n\right].$$

$$= \frac{4}{\pi^2 n^2}\left[1-(-1)^n\right].$$

if n is even.

$$\frac{4}{\pi^2 n^2}[1-1] = 0 \Rightarrow 0$$

if n is odd.

$$\frac{4}{\pi^2 n^2}[1+1] = \boxed{\frac{8}{\pi^2 n^2}}$$

Q.No.

$$bn = \frac{1}{\pi}\int_{-0}^{\pi} f(x) \sin nx dx.$$

where,

$$bn = \frac{1}{\pi}\int_{-\pi}^{0}\left(1+\frac{2x}{\pi}\right)\sin nx dx + \int_{0}^{\pi}\left(1-\frac{2x}{\pi}\right)\sin nx dx.$$

$$= \frac{1}{\pi}\left[\left(1+\frac{2x}{\pi}\right)\left(\frac{-\cos nx}{n}\right) - \left(\frac{2}{\pi n}\right)\left(\frac{\sin nx}{n}\right)\right]_{-\pi}^{0}$$

$$+ \left[\left(1-\frac{2x}{\pi}\right)\left(\frac{-\cos nx}{n}\right) - \left(\frac{-2}{\pi n}\right)\left(\frac{\sin nx}{n}\right)\right]_{0}^{\pi}.$$

$$= \frac{1}{\pi}\left[(0)-(0)\frac{2}{\pi n^2}\left(-1 (-1)^n\right) - \frac{-2}{\pi n^2}\left(1(-1)^2\right)(0)(0)\right]$$

$$= \frac{1}{\pi}\left[\frac{2}{\pi n^2}\left(1(-1)^n\right) - \frac{2}{\pi n^2}\left(1 (-1)^2\right)\right]$$

Q.No.

$= \frac{1}{\pi} [0]$

$= 0.$

$\therefore \boxed{b_n = 0}$

Substituting the values in the Fourier series expansion.

$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \int_0^{\pi} f(x) \cos nx + \sin nx$

$f x = 0 + \sum_{n=1}^{\infty} \int_0^{\pi} \frac{8}{\pi^2 n^2} + 0$

$$\boxed{f x = \sum_{n=1}^{\infty} \int_0^{\pi} \frac{8}{\pi^2 n^2}}$$

Q.No.

1b)

given that

The Fourier Series of

$f(x) = x \cos x$

In the interval $(-\pi, \pi)$.

The fourier series expansion

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \int_{-\pi}^{\pi} a_n \cos nx + b_n \sin nx.$$

where,

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\, dx.$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cdot \cos nx\, dx.$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx\, dx.$$

Q.No.

To find ao

$$\Rightarrow a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} F(x)\,dx.$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} (x \cos x)\,dx.$$

$$= \frac{1}{\pi}\left[x \cdot \frac{\sin nx}{n} + x \cdot \frac{\cos nx}{n}\right]_{-\pi}^{\pi}.$$

$$= \frac{1}{\pi}\left[\pi \cdot \frac{\sin n\pi}{n} + (-\pi)\frac{\cos n\pi}{n}\right].$$

$$= \frac{1}{\pi}\left[\frac{\pi}{n} + \frac{-\pi}{n}\right]$$

$$= \frac{1}{\pi}[0]$$

$$= \frac{1}{\pi}[0] = 0$$

$$\therefore \boxed{a_n = 0}$$

Q.No.

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi}(x\cos x)\cos nx\,dx.$$

$$= \frac{1}{\pi}\left[x\cdot\frac{\cos nx}{n}\cdot\frac{1}{x}\cdot\frac{\sin nx}{n}\right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi}\left[\pi\frac{\cos n x}{n}\cdot\frac{1}{-\pi}\frac{\sin nx}{n}\right].$$

$$= \frac{1}{\pi}\left[\pi - \pi\right]$$

$$= \frac{1}{\pi}[0]$$

$$= 0$$

$$\boxed{a_n = 0}$$

Q.No.

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) \sin nx \, dx.$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} (x\cos x) \sin x \, dx.$$

$$= \frac{1}{\pi}\left[x \cdot \frac{\sin nx}{n} \cdot \frac{1}{n} - \frac{\cos nx}{n}\right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi}\left[x \cdot \frac{\sin nx}{n} \cdot \frac{1}{n} - \frac{\cos n}{n}\right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi}\left[\pi \cdot \frac{\sin n\pi}{n} \cdot \frac{1}{n} - \frac{\cos n\pi}{n}\right]$$

$$= \frac{1}{\pi}\left[-\pi + \pi - \pi\right].$$

$$= \frac{1}{\pi}[0]$$

$$= \frac{1}{\pi}[0] = 0$$

$b_n = 0.$

$\boxed{a_0 = 0}$ $\boxed{a_n = 0}$ $\boxed{b_n = 0}$

Q.No.

from the fourier Series Expansion.

$$\boxed{f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \int_{-\pi}^{\pi} \cos nx + \sin nx}$$

Q.No.

3b)

given that

$F(x) = 1$
$f(x) = 0$

The given limits = $(0, 2)$
$(x, 2)$.

To find sine and cosine transforms.

To find sine transform

Where the expansion of sine transform

$$\boxed{F(x) = \sum_{n=1}^{\infty} \int_0^1 b_n \sin n x x} \quad b_n = \frac{2}{\pi}\int_0^1 F(x)\sin n x dx$$

Where, in the sine transform.

$$\boxed{\begin{aligned} a_0 &= 0 \\ a_n &= 0 \end{aligned}}$$

Q.No.

To find the cosine transform

The Fourier of transform of cosine

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \int_0^1 a_n \cos nx$$

where,

$$a_0 = \frac{2}{\pi} \int_0^1 f(x)\, dx.$$

<u>To find an</u>

$$a_n = \frac{2}{\pi} \int_0^1 f(x) \cos nx\, dx$$

where,

$$\boxed{b_n = 0}$$

Q.No.

7a,

<u>given that</u>

The inverse laplace transform

$$L^{-1}\left\{\frac{2s-3}{s^2+4s+13}\right\}$$

The laplace transform formulas

$$\boxed{\begin{aligned} L(1) &= \frac{1}{s} \\ L(t^n) &= \frac{n!}{(s)^{n+1}} \\ L(\sin bt) &= \frac{b}{s^2+b^2} \\ L(\cos bt) &= \frac{s}{s^2+b^2} \\ L(\sinh bt) &= \frac{b}{s^2-b^2} \\ L(\cosh bt) &= \frac{s}{s^2-b^2} \end{aligned}}$$

Q.No.

3(a)

given that,

The Fourier integral

show that

given eq $\int_0^\infty \frac{\cos \lambda x}{1+\lambda^2} d\lambda = \frac{\pi}{2} e^{-x}; x \geq 0$

where
$$f(x) = \int_0^\infty \frac{\cos \lambda x}{1+\lambda^2} d\lambda.$$

$$\text{limits} = \frac{\pi}{2} e^{-x} \geq 0.$$

The Fourier integral.

$$a_0 = \frac{2}{\pi} \int_{\frac{\pi}{2}}^{0} f(x)\, dx.$$

$$a_n = \frac{2}{\pi} \int_{\frac{\pi}{2}}^{0} f(x) \cos nx\, dx.$$

$$b_n = \frac{2}{\pi} \int_{\pi}^{0} f(x) \sin nx\, dx.$$

SEMI-LOG PAPER (5 CYCLES X 1/10")